MIL-C-24643/25B
3 October 1986
SUPERSEDING
MIL-C-0024643/25A(SH)
8 November 1985
MIL-C-24643/25
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LSPI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight with circuit integrity)

First	- Copper conductor, nickel coated, AWG 18, class B, stranding.
Second	- Extruded silicone rubber insulation. Nominal thickness 0.018 inch.
Third	- Glass braid (one colored black and one colored white for each pair of conductors).
Fourth	- Braid covering. Two conductors (one black, one white), cabled together with a lay not greater than 2-1/4 inches, to form a pair.
Fifth	- Braided shield of AWG no. 34 or no. 36 uncoated copper over each pair. Minimum braid angle 30 degrees; minimum coverage 85 percent.
Sixth	- Shield insulation over each braided shield, consisting of polyethylene terephthalate film, type G in accordance with MIL-I-631, 0.001-inch thick, applied helically with 50-percent overlap.
Seventh	- A glass braid over each shield insulation. (Standard identification code applied by method 4).
Eighth	- The specified number of pairs (see table I) cabled together with a lay not greater than 14 inches. Fillers saturated with noncharring silicone-base compound shall be used as necessary to make a firm, round cable.

AMSC N/A FSC 6145
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Ninth - Binder tape applied helically with overlap.
Tenth - Silicone rubber jacket (see table I for thickness).
Eleventh - Braided metal armor.
Twelfth - Paint.

TABLE I. Details.

Military part no. M24643/25	Type and size	Number of pairs	Cable jacket thickness minimum (inch)	Overall diameter maximum (inch)	Cold bending mandrel diameter maximum (inches)
-01AN	LSPI-3	3	0.040	0.685	9.0
-02AN	LSPI-7	7	.050	.900	11.0
-03AN	LSPI-12	12	.050	1.155	14.0

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet at 25°C maximum 1/.	7.5
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	1200
Conductor to shield	700
Shield to shield	500
Insulation resistance - megohms/1000 feet, minimum	500
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
The nickel coating on the conductors of the completed cable shall not be nicked, scraped, or abraded as to expose the copper conductor	No failure
Group B:	
Cold bending, cable (see table I for mandrel diameter)	No damage
Gas flame, 1 hour at 120/70 volts root mean square	No failure
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum	750
Elongation - percent, minimum	125
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	800
Elongation - percent, minimum	250
Durometer (type A) hardness, minimum	60

1/ Resistance shall be measured on 100 percent of conductors; values shall be included in a certified test report.

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Group C:
Physicals - cable aging (260 ± 10°C)
Jacket

Tensile strength - lb/in^2, minimum.....	600
Elongation - percent, minimum.....	150

QUALIFICATION INSPECTION:
Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Requirements
Cable aging and compatibility (125 ± 5°C) - for this test, omit all references to watertightness and fillers).....	No failure
Gas flame (3 hours) at 120/70 volts root mean square....	No failure
Armor - conformance to material, construction and coverage.....	No failure
Specific gravity of extruded silicone insulation, maximum.....	1.55
Flame propagation.....	No failure

UNIT ORDERING LENGTHS:
All types and sizes: 100 feet minimum

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1005-25)

Review activities:
Army - AV, CR, ER
Navy - EC
DLS - IS

User activities:
Army - ME, AR, AL
Navy - CG

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